Structure of the Paper

Title

Abstract, keywords

1. Introduction

2. Literature review & Major contributions in 5 points

3. Dataset overview & Augmentation (add dataset class-wise description table, dataset sample images)

4. Description of proposed model architecture (add overview diagram)

5. Results and Discussion (add all plots, big performance metrics table of your proposed model)

6. Ablation study (add one comparison table)

7. Qualitative comparative analysis of proposed model wrt existing models (add mAP vs Parameter plot in Origin Pro)

8. Deployment on edge device (Raspberry Pi specs, add setup photo)

9. Conclusion

10. References

Note: But do not start writing the paper in the above order. First, begin with the most important section of your paper, which is Section 5: Results and Discussion. Reviewers are primarily interested in the obtained results and their explanations; therefore, this section must be written very effectively. Next, write Section 6: Ablation Study, followed by Section 8: Hardware Deployment.

You have to strictly follow this order while writing the paper:

Section 5. Results and Discussion

Section 6. Ablation study

Section 8. Deployment on edge device

Section 7. Qualitative comparative analysis

Section 4. Description of proposed model architecture

Section 3. Dataset overview & Augmentation

Section 2. Literature review

Section 1. Introduction

Finally write these: major contributions, frame a good title, write abstract and conclusion

Check whether all reference citations given in Literature review, Qualitative comparative analysis and introduction sections.

Begin writing the content from here

Suggestion: Keep the table (before & after augmentation, hybridisation without AI & hybridization with AI results) with all performance metrics (mAP 0.5, mAP 50-95, Parameters, Precision, Recall, inference time, GFLOPs) ready.

Note: First open a new chat (better use chatgpt 5.2 or better version) and give the dataset description first. You have to go to the source from where you have collected the dataset for dataset description.

Dataset description:

Prompt: Analyze the dataset description in detail. Analyze the distinct classes available in the dataset and also analyse if class imbalance etc.

Note: Next, let know what model you are using.

Prompt: I have used Yolov5s for object detection and this is the yaml code (give the model summary/ yaml file code):

Note: Next, you have to write the Results and Discussion section. Mainly focus on the results (performance metrics) you obtained, why you obtained these results, and the in-depth reasons behind your model producing such outcomes.

Section. 5 Results & Discussion

Note: Give brief description of your local workstation specifications, training splits, Python version, framework details etc.

A. **Performance analysis Before Augmentation**: **(1 paragraph 250 words)**

Prompt: These are the results (confusion matrix and Precision-recall curve) obtained by the baseline YOLOv11n model (give yaml code). The obtained mAP0.5 is 5.7 and the parameter count is 2.56Million. Perform class-wise analysis & which are the poor performing classes and best performing classes and give the in-depth reasons why & how the model has given such overall mAP0.5. What component/blocks of the baseline YOLOv11n model has contributed to the obtained mAP0.5 and also why the model performed poor in some classes.

B. **Performance analysis After augmentation (1 paragraph 150 words)**

Prompt: We have augmented the dataset using ..... techniques (give augmentation code).

Prompt: These are the results (confusion matrix and Precision-recall curve) obtained by the baseline YOLOv11n model after augmentation. The obtained mAP0.5 is 98.7 and the parameter count is 2.56Million. Perform class-wise analysis & give the in-depth reasons why & how the model has given such overall mAP0.5 after augmentation. Analyze how the augmentation techniques have contributed to the enhanced mAP0.5 compared to mAP before augmentation.

C. **Performance analysis After hybridization (without AI domain included) (1 paragraph 300 words)**

Note: We have to discuss only about the proposed optimal (trade-off) configuration which we have obtained among the 5 to 6 configurations you have put in ablation study. Now you have to give the code of that proposed model. And frame a good name for your proposed model (based on the main impactful blocks in ur model, something like this: DA-ActNN-YOLOV5…….)

Prompt: I have performed the architectural changes in the baseline Yolov11n model as per this yaml code. This is the hybridized model (give Yaml code). Analyze the hybridized architecture of the model. These are the results (confusion matrix and Precision-recall curve) obtained by the hybridized model. The obtained mAP0.5 is 98.7 and the parameter count is 1.3 Million. Perform class-wise analysis & give the in-depth reasons why & how the model parameters count has been reduced significantly while achieving the optional mAP0.5 after hybridization along with the contribution of the AI domain used. Analyze what blocks contributed in making the model lightweight with lowest parameter count while achieving the optional mAP0.5.

D. **Performance analysis After hybridization (with AI domain included) (1 paragraph 300 words)**

This is the hybridized model with the AI domain included. Analyze how AI Domain has been included in the overall framework. These are the results (confusion matrix and Precision-recall curve) obtained by the hybridized model. The obtained mAP0.5 is 98.7 and the parameter count is 1.3 Million. Perform class-wise analysis & give the in-depth reasons why & how the integration of AI domain has impacts the results obtained.

E. **Performance metrics Analysis (1 paragraph 200 words)**

Prompt: Further analyze the impact of the hybridized model on the Parameter count, inference time, and GFLOPs. Highlight the suitability of the lightweight proposed model for edge deployment.

Note: Add Confusion Matrix, Precision vs recall plot, Predictions labels, Labelling and bounding box positions & training and validation losses of each sub-section above

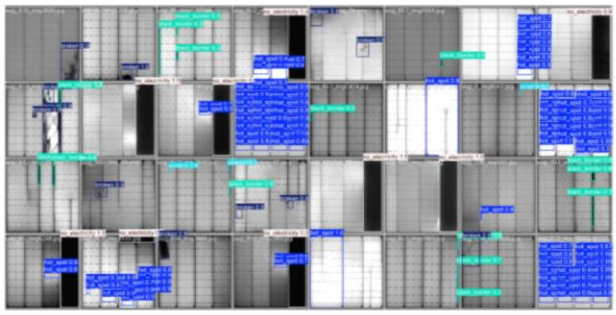


Fig. 5. Predictions of defect detections of PV-multi defect dataset.

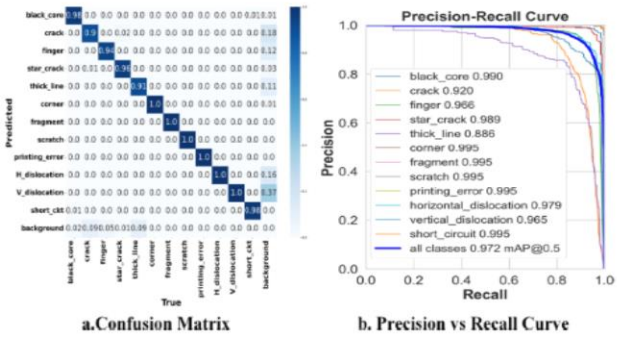


Fig. 6. (a) Confusion Matrix and (b) Precision vs recall plot obtained for PVEL-AD dataset.

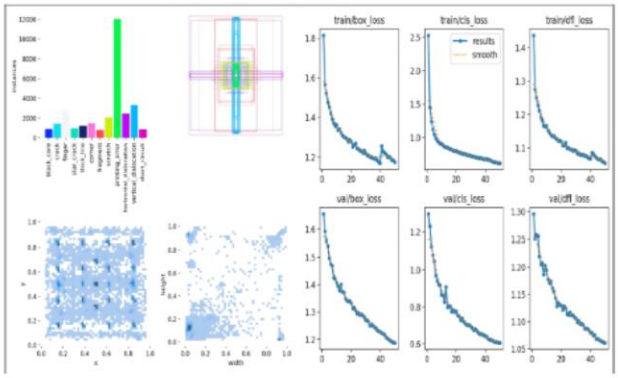


Fig. 7. Labelling and bounding box positions & training and validation losses obtained for PVEL-AD dataset.

salient regions, improving discrimination between similar defect types. These improvements are supported by the confusion matrix and precision-recall curves (Fig. 9(a)). Detection examples (Fig. 5) also confirm the model's strong localization and classification performance under diverse conditions. DCGAN-based augmentation was again essential. The generator was designed to synthesize class-conditional defect patches by appending one-hot vectors, enabling support for rare defects (e.g., fine parasite-induced etching) with

Section. 6 Ablation Study

Ablation study **(2-3 paragraph 650 words)**

Prompt: I want to perform the ablation study. I have tried 5 different configurations. Now i will give different configurations analyze them one by one, their impact on the obtained mAP0.5 and the parameter count how this configuration & what component of this configuration has resulted in the obtained mAP0.5 and parameter count.

Prompt: This is the code of configuration-1 'give yaml code'. Give one specific name of this particular model. Analyse it's impact on mAP and parameter count.

Prompt: This is the code of configuration-2 'give yaml code'. Give one specific name of this particular model. Analyse it's impact on mAP and parameter count.

Prompt: This is the code of configuration-3 'give yaml code'. Give one specific name of this particular model. Analyse it's impact on mAP and parameter count.

Prompt: This is the code of configuration-4 'give yaml code'. Give one specific name of this particular model. Analyse it's impact on mAP and parameter count.

Prompt: This is the code of configuration-5 'give yaml code'. Give one specific name of this particular model. Analyse it's impact on mAP and parameter count.

Note: Keep the table (all the different configurations u tried) with all performance metrics (mAP 0.5, mAP 50-95, Parameters, Precision, Recall, inference time, GFLOPs) ready.

Prompt: This is the table 'give the table' consisting of 5 different configurations that I have tried with the obtained performance metrics. Among the 5 configurations, the fifth confirmation is considered as the best one due to its balance trade off solution between the mAP0.5 and parameter count.

Note: Add a table in paper at the end of this section showing all configurations and their performance metrics.

TABLE III
ABLATION STUDY RESULTS COMPARING DIFFERENT MODEL CONFIGURATIONS

Model Config.	Param(M)	GFLOPs	mAP50(%)	mAP50-95(%)	Recall	Precision
C2f Baseline	2.26	7.3	95.9	82.1	0.917	0.968
Adown+C3x+Ghost+RepC3+C3k2	1.80	13.1	95.2	78.7	0.913	0.959
C3Ghost+CBAM	1.88	9.0	93.0	73.9	0.905	0.896
C3Ghost+DWConv	1.54	7.4	91.8	77.3	0.884	0.908
C3Ghost	1.90	6.0	93.4	72.9	0.878	0.938
Downscale + Upscale (Proposed)	**1.27**	**4.5**	**95.3**	**77.07**	**0.906**	**0.952**

Section.7 Qualitative comparative analysis

Note: Keep the table (results of all different models from 4-7 existing papers) with all performance metrics (mAP 0.5, mAP 50-95, Parameters, Precision, Recall, inference time, GFLOPs) ready.

A. Qualitative & Quantitative comparative analysis (2-3 paragraphs 650 words)

Prompt: Perform a detailed comparative analysis of the proposed model with the existing models. Highlight and analyse how the proposed model/framework is highly effective in mAP with very lightweight compared to each existing model. Give a brief paragraph for each model comparison. Include the citation references of the existing models in table as well as in discussion.

Note: Add a table in paper at the end of this section showing all existing model performance metrics along with the reference citations.

TABLE IV
PERFORMANCE METRICS COMPARISON OF VARIOUS MODELS ON PVEL-AD AND PV-MULTI-DEFECT DATASETS

Ref	MODEL	mAP50	mAP50-95	PRECISION	RECALL	PARAMETERS (M)	INFERENCE (ms)	GFLOPS
				On PVEL-AD Dataset				
[14]	YOLOV5s	81.9	-	-	-	7.1	-	-
[7]	YOLOv8n	90.38	62.31	-	-	5.95	4.3	5.95
[5]	YOLO-ACF	92.6	62.1	84.5	92.3	6.1	6.8	14.2
[3]	YOLOv8-l	93.96	67.89	67.90	75.09	43.63	16.52	82.71
[4]	Yolov8n (MLCA)	91.5	-	-	-	3.01	-	8.1
[5]	YOLO v3	60.5	26.9	53.1	72.4	8.7	7.3	12.9
[4]	Yolov7tiny	50.8	-	-	-	6.04	-	13.3
[13]	YOLOV5-l	81.93	51.13	-	-	50.3	-	-
[7]	Retina Net(C2DEM)	74.30	45.50	-	-	139.0	31.9	18.8
[5]	RetinaNet	91.3	58.7	69.7	58.8	10.4	37.4	68.1
[14]	EFFICIENTDET-D0	78.7	-	-	-	3.9	-	-
[4]	Yolov8m	91.7	-	-	-	25.85	-	78.1
[7]	C2DEM-YOLO	92.31	63.55	-	-	7.9	7.7	7.9
[14]	PSA-YOLOV7	82.9	-	-	-	37.5	-	-
[4]	Yolov8s-C2f MLCA	92.3	-	-	-	11.13	-	28.5
[5]	YOLO v6	75.7	51.7	67.6	77.1	16.3	9.1	44.0
	YOLOv11n+DCGAN	**95.0**	**71.8**	**92.2**	**95.9**	**2.58**	**3.0**	**6.3**
	YOLOv11n-GhostLite + DCGAN	**97.2**	**83.4**	**93.7**	**96**	**2.34**	**2.2**	**6.2**

Section.8 Hardware Deployment

Deployment on Edge computing devices.

You can put any one of the following section headings for this particular section. Explore for any other good headings also.

- Edge Hardware Implementation and Validation on Embedded Platforms
- Hardware Validation and Edge-AI Deployment Analysis
- Real-Time Edge-AI Deployment and Hardware Validation on Embedded Devices
- Experimental Hardware Validation on Resource-Constrained Edge Platforms
- Edge-AI Hardware Validation and Real-Time Deployment Performance
- Real-Time Edge-AI Deployment and Hardware Performance Evaluation
- Implementation and Hardware Validation on Edge Platforms

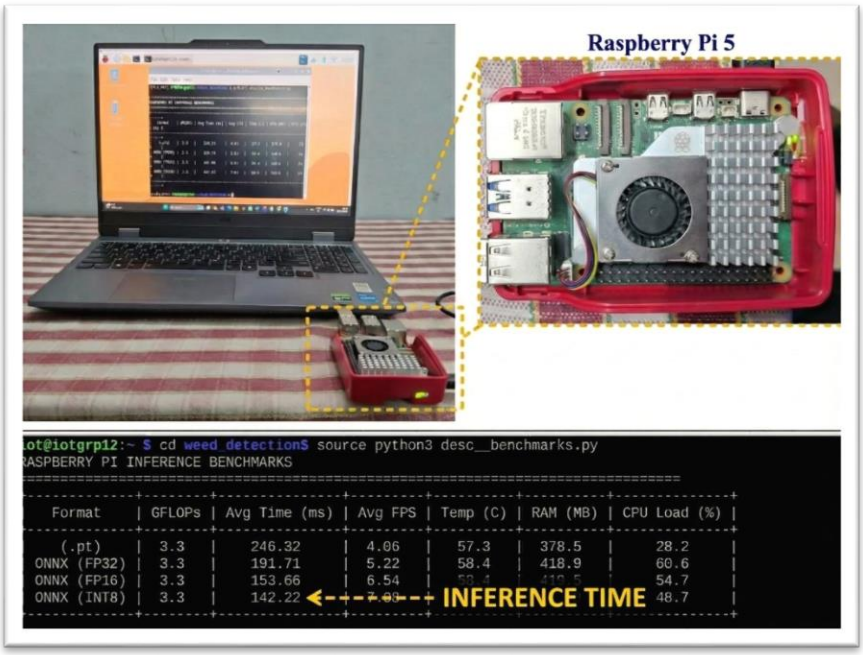
The following subsections should be included in this section:

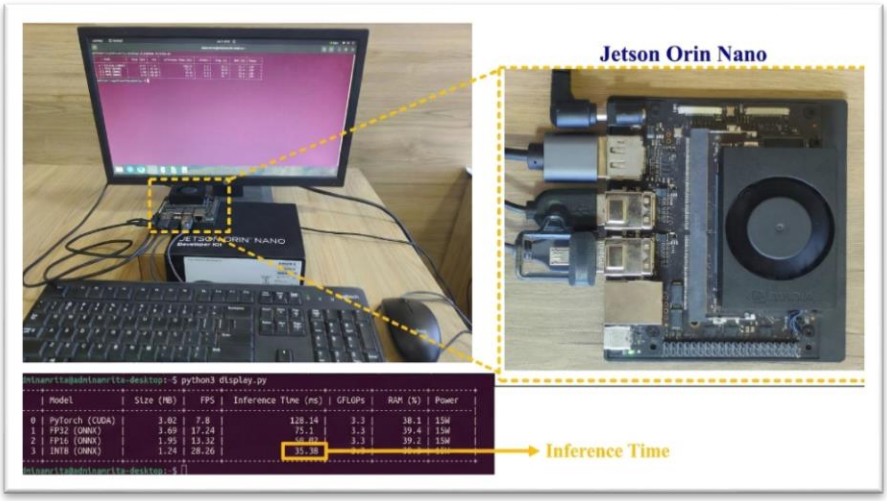
- A. Experimental Setup and Hardware Specifications (of both the boards)
- B. Edge Deployment on Raspberry Pi
- C. Edge Deployment on NVIDIA Jetson Orin Nano (discuss about inference latency across PyTorch (CUDA), FP32, FP16, and INT8 ONNX execution modes)
- D. Comparative Performance Analysis

Prompt: These are the results (give the screenshot of inference time) obtained by deploying the proposed model on Raspberry Pi/NVIDIA Jetson Orin Nano. Perform the analysis of the inference time obtained by the proposed model and also highlight how the lightweight nature of the proposed model yielded the fast inference time.

Prompt: Write one brief paragraph on inference time on Raspberry Pi (in 100 words) and a paragraph on inference time on NVIDIA Jetson Orin Nano (in 200 words) and the comparative performance analysis of raspberry pi/NVIDIA Jetson Orin Nano (in 150 words).

Add the overall setup diagrams, specifications table of Raspberry & Jetson, inference metrics in table.





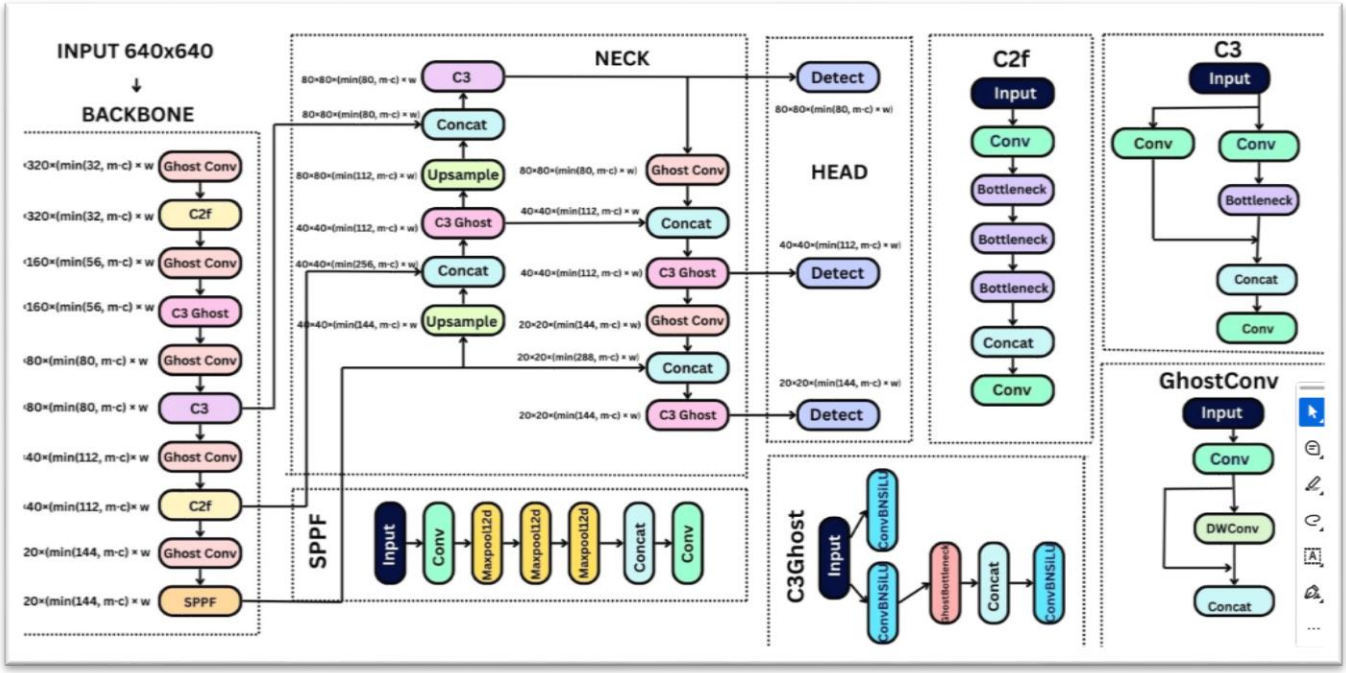
Section-4 Proposed Framework

Proposed YOLO model architecture

Prompt: This is the yaml code of the proposed model. Give the description of the proposed model architecture in detail with strong mathematical rigor. Give in paragraphs with side headings and with strong mathematical rigor. Emphasis the description of the specialized blocks used in the architecture. Finally highlight how this proposed architecture is different and efficient from the baseline model.

Prompt: Give the description of the AI domain (write the name of AI domain used) used in the proposed pipeline. This is the code 'insert the code'. Give the description and the usage of the AI domain (write the name of AI domain used) used in the proposed pipeline in detail with strong mathematical rigor.

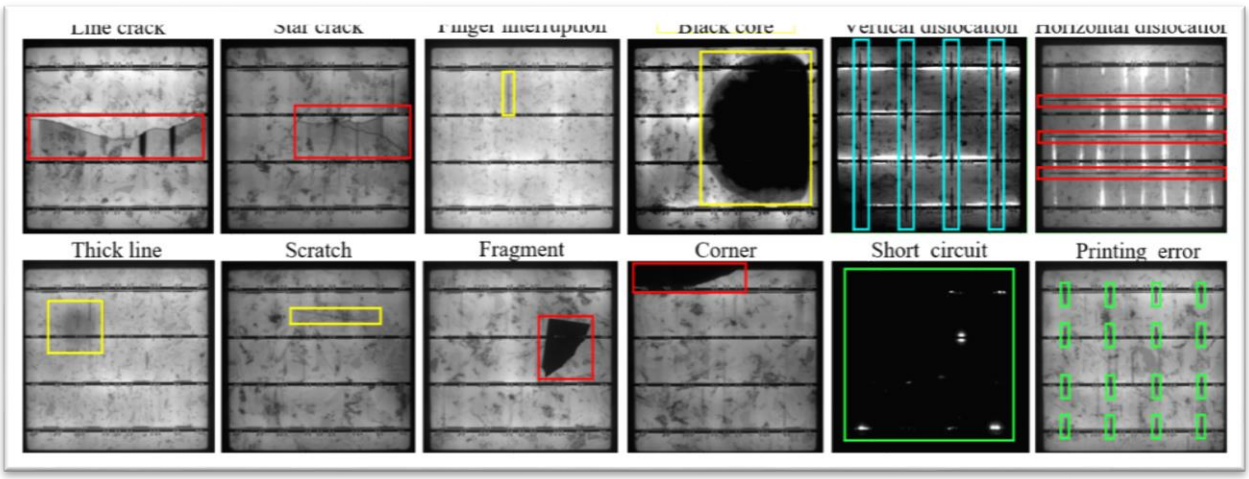
- Draw the overall framework with AI domain included.
- Draw the detailed hybridized model architecture with the internal diagrams of specialized blocks.



Section-3 Dataset Overview and Augmentation

Prompt: Give the description of the dataset in a paragraph.

Put some sample images & various classes in table in the paper.



Prompt: We have augmented the dataset using ..... techniques (give augmentation techniques code). Give the description of the augmentation techniques used with strong mathematical rigor and equations

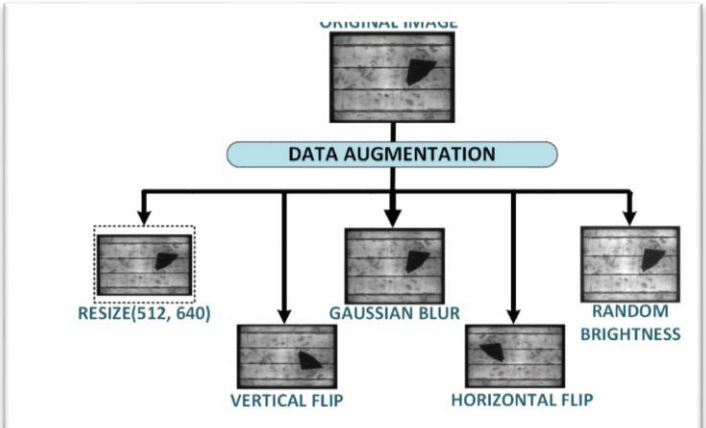


Fig. 3. Sample images after augmentation on PVEL-AD dataset.

Section-2 Literature Review

Literature Review: Description of work done, model used, advantages & limitations (wrt proposed work).

Take 16-18 related (only Q1-Q3) journal papers from google scholar (papers used your dataset should be must included).

Prompt: Give the description of work done, model used, advantages & limitations (wrt my proposed model) for Paper-1.

Give the description of work done, model used, advantages & limitations (wrt my proposed model) for Paper-2.

Give the description of work done, model used, advantages & limitations (wrt my proposed model) for Paper-3.

Give the description of work done, model used, advantages & limitations (wrt my proposed model) for Paper-4……till paper-18

Gather all the content from these 18 papers for framing the literature review.

Prompt: This is the literature review content gathered from 18 papers. Give the literature review in a table format mentioning the refrence & year, description of work done, model used, advantages & limitations (wrt proposed work). Include the reference citations also for each paper.

TABLE I: Comparative Analysis of Existing PV Defect Detection Methods

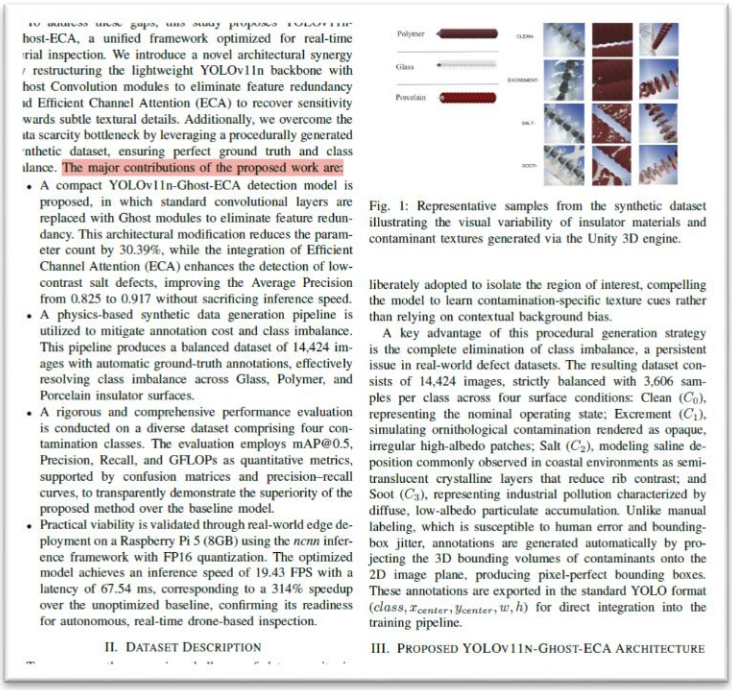
Paper	Description	Model	Advantages	Limitations
[1] 2020	Scalable object detection with compound scaling	EfficientDet D0-D7 + BiFPN	Excellent accuracy-efficiency trade-off; highly scalable	Performance drops on very small defects; BiFPN compute-heavy
[2] 2023	Anomaly detection with multi-channel spatial attention	CNN + MC-Spatial Attention	Lightweight; good anomaly classification	No localization capability; limited multi-scale reasoning
[3] 2024	Lightweight CNN for EL defect detection	Shallow custom CNN	Fast inference; simple architecture	Limited feature depth; weaker small-crack detection
[4] 2024	YOLO-ACF with adaptive complementary fusion	YOLO + ACF attention	Strong small-defect detection; better fusion	Attention increases computational cost
[5] 2025	Channel & Coordinate Aware YOLO	YOLO + CCA attention	Improved localization; robust under noise	Higher memory use; increased latency
[6] 2025	Enhanced YOLOv8 with dual effective modules	YOLOv8 + C2f-DE + EM	High recall on cracks; deep extraction	Heavy architecture; unsuitable for embedded devices
[7] 2021	Complementary attention network	CNN + attention classifier	Good noise robustness; strong metrics	No detection head; cannot localize cracks
[8] 2024	RAFBSD with shallow-deep fusion	CNN with residual-attention	Balanced speed and accuracy	Limited sensitivity for tiny crack patterns
[9] 2024	Improved YOLOv7 for PV EL	YOLOv7 with modified anchors	High accuracy; strong EL crack detection	YOLOv7 remains heavy; high resource needs

Section-1 Introduction

Prompt: Give the generic introduction & background of the research problem. How different methods have been used to solve the problem? And how AI and Deep learning and yolo has been instrumental in solving it. And still what are the problems with Yolo and hence what is the need/motivation for the proposed framework.

Major contributions- put it at the end of literature review

Prompt: Give the major contributions of the overall work in 5 brief paragraphs.



To address these gaps, this study proposes YOLOv11n-Ghost-ECA, a unified framework optimized for real-time industrial inspection. We introduce a novel architectural synergy by restructuring the lightweight YOLOv11n backbone with Ghost Convolution modules to eliminate feature redundancy and Efficient Channel Attention (ECA) to recover sensitivity towards subtle textural details. Additionally, we overcome the data scarcity bottleneck by leveraging a procedurally generated synthetic dataset, ensuring perfect ground truth and class balance. The major contributions of the proposed work are:

- A compact YOLOv11n-Ghost-ECA detection model is proposed, in which standard convolutional layers are replaced with Ghost modules to eliminate feature redundancy. This architectural modification reduces the parameter count by 30.39%, while the integration of Efficient Channel Attention (ECA) enhances the detection of low-contrast salt defects, improving the Average Precision from 0.825 to 0.917 without sacrificing inference speed.
- A physics-based synthetic data generation pipeline is utilized to mitigate annotation cost and class imbalance. This pipeline produces a balanced dataset of 14,424 images with automatic ground-truth annotations, effectively resolving class imbalance across Glass, Polymer, and Porcelain insulator surfaces.
- A rigorous and comprehensive performance evaluation is conducted on a diverse dataset comprising four contamination classes. The evaluation employs mAP@0.5, Precision, Recall, and GFLOPs as quantitative metrics, supported by confusion matrices and precision–recall curves, to transparently demonstrate the superiority of the proposed method over the baseline model.
- Practical viability is validated through real-world edge deployment on a Raspberry Pi 5 (8GB) using the *ncnn* inference framework with FP16 quantization. The optimized model achieves an inference speed of 19.43 FPS with a latency of 67.54 ms, corresponding to a 314% speedup over the unoptimized baseline, confirming its readiness for autonomous, real-time drone-based inspection.

II. Dataset Description

Fig. 1: Representative samples from the synthetic dataset illustrating the visual variability of insulator materials and contaminant textures generated via the Unity 3D engine.

liberately adopted to isolate the region of interest, compelling the model to learn contamination-specific texture cues rather than relying on contextual background bias.

A key advantage of this procedural generation strategy is the complete elimination of class imbalance, a persistent issue in real-world defect datasets. The resulting dataset consists of 14,424 images, strictly balanced with 3,606 samples per class across four surface conditions: Clean (C_0), representing the nominal operating state; Excrement (C_1), simulating ornithological contamination rendered as opaque, irregular high-albedo patches; Salt (C_2), modeling saline deposition commonly observed in coastal environments as semi-translucent crystalline layers that reduce rib contrast; and Soot (C_3), representing industrial pollution characterized by diffuse, low-albedo particulate accumulation. Unlike manual labeling, which is susceptible to human error and bounding-box jitter, annotations are generated automatically by projecting the 3D bounding volumes of contaminants onto the 2D image plane, producing pixel-perfect bounding boxes. These annotations are exported in the standard YOLO format $(class, x_{center}, y_{center}, w, h)$ for direct integration into the training pipeline.

III. Proposed YOLOv11n-Ghost-ECA Architecture

Abstract, conclusion and Title.

Prompt: Give the abstract section of the paper (in one paragraph, 200 words)

Prompt: Give the conclusion section of the paper (in one paragraph, 225 words)